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The scheme for globalization of a process-based model explaining gradations in terrestrial NPP and its application

G.A. Alexandrov ^{a,*}, T. Oikawa ^b, Y. Yamagata ^a

^a *National Institute for Environmental Studies (NIES), Tsukuba, Ibaraki 305-0053, Japan*

^b *Institute of Biological Science, Tsukuba University, Tsukuba, Ibaraki 305-8572, Japan*

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Abstract

It is more or less obvious that a process-based net primary production (NPP) model tested for a stand can be applied over a global grid, but doing so, we face the problem—how to calibrate it at this scale. In this paper we demonstrate a scheme for such calibration with a process-based NPP model. First, we suggest that model parameters should be derived from a freely accessible database of NPP measurements. Then, we reduce the number of undefined model parameters to a single lumped parameter (SLP) and find its value for each database entry. Next, considering SLP as some characteristic of vegetation, we formulate an empirical model linking SLP and climate. And finally, we globalize the process-based model in hand by assigning SLP value to the nodes of geographical grid proceeding from the global fields of climatic variables. © 2002 Elsevier Science B.V. All rights reserved.

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1. Introduction

Terrestrial net primary production (NPP) has been a focus of biosphere studies over the last three decades. Common methodology of NPP measurements was elaborated in course of International Biological Programme (IBP) from 1965 to 1974. Then, in course of International Geosphere–Biosphere Programme (IGBP), which was started in 1990, the bulk of NPP data collected

during IBP has been turned into a number of NPP models. This has transformed general knowledge about the potential productivity of plants into a quantitative theory.

However, the predictive power of this theory at the global scale is questionable. Since NPP at this scale cannot be observed directly, its magnitude is assessed by applying a model, calibrated from a limited set of field and laboratory studies, over the geographical grid. The result is obviously dependent on how the model was calibrated, but most of the recent publications on global carbon cycle modelling avoid this important question. More-

* Corresponding author. Tel./fax: + 81-298-512-960.

E-mail address: g.alexandrov@nies.go.jp (G.A. Alexandrov).

over, Cramer and Field (1999) advocate the idea that there is no way (at least at the present) in which the behavior of a global scale model can be evaluated except by comparison to other models of this sort (Cramer et al., 1999). All this cast some doubts on the credibility of the global scale NPP models: in which sense and for which purpose they could be used, if they could not be supported by the data?

The goal of this paper is to evaluate the existing potential for empirical validation of global scale NPP models and come up with a process-based model transparently linked to the NPP data. In the first place, we discuss the validity of available NPP data and explain our interpretation of these data in terms of relationship between biome-averaged NPP and climate. Secondly, we consider the ways in which NPP models are calibrated and propose a scheme for globalization of a process-based NPP model. Finally, we apply this scheme to the NPP model proposed by Oikawa (1986) and discuss the predictive ability of thus obtained global scale NPP model.

2. The scheme for globalization of process-based model

The words ‘model is valid’ may mean either ‘consistent with available data’ or ‘in accord with current knowledge and beliefs’. A model that satisfies both of the validation criteria is simply referred to as valid (see Rykiel (1996) for the detailed review of the terminology). If model structure, the list of variables and causal relationships between them, provide reasonable representation of the real system, the model is said to be *conceptually valid*.

A conceptually valid model does not necessarily fit available data. The quality of the data is limited by the technique of measurements, and so it is not always possible to assume that the data accurately represent the real system. In this case, one may argue that a conceptually valid model provides better representation of reality. Hence, empirical validation of a conceptually valid model should be started from the validation of the observed data.

2.1. Reference data set

The Distributed Active Archive Center (DAAC) of the Oak Ridge National Laboratory (ORNL) maintains and distributes an extensive database on NPP (Esser et al., 1997). This database was originally synthesized by a group of scientists from the University of Osnabrück (Germany) led by H. Lieth in 70–80s. It includes over 700 estimates of NPP extracted from pre-82 publications (reflecting mainly IBP activity). Although, this database has not been updated since that time, it contains fundamental facts for estimating terrestrial NPP.

That collection of NPP data was not produced by a random choice of data points, and hence one needs additional classifying information to analyze the data properly. Proceeding from this information, Alexandrov et al. (1999) excluded some records and thus came to a reference data set that meets following criteria:

1. contains only original estimates of total NPP (i.e. estimate of total NPP reported directly by primary data source rather than re-calculated from primary data in course of database compilation);
2. geographical location of data point allows to specify vegetation zone, mean annual temperature and precipitation;
3. shows normal distribution of NPP within the groups related to vegetation zones.

We complemented this data set with the data on mean annual temperature (T_a), annual precipitation (P_a), actual evapotranspiration (AET), length of the growing season (G) and its average light intensity (I_0), temperature (T_v), and precipitation (P_v). AET values were taken from the TGER-database (Box et al., 1994). The data set of Leemans and Cramer (1991) was used to calculate T_a , P_a , T_v and P_v . The global field of I_0 was derived from the data set on PAR (Dye et al., 1994). Geographical distribution of G was derived from the characteristic month averages (1986–1988) of the monthly NDVI (Kinemann and Hastings, 1992) and T_m , monthly mean temperatures (Leemans and Cramer, 1991), by use of $T_m > 0$ °C and NDVI > 0.1 as a criterion for growing season (see Myneni et al., 1997 for dis-

cussion of NDVI-based criteria for growing season).

Finally, following the idea (Lieth, 1975) that NPP of potential natural vegetation is limited with certain value, we excluded records with NPP higher than $1350 \text{ g C m}^{-2} \text{ per year}$.

2.1.1. Consistency with current beliefs

The reference data set complemented with the climate data provides us the estimate of the biome-averaged NPP as well as characteristics of its average climate. Plotting NPP against a climate characteristic allows us to see the shape of the curve describing climate dependence of NPP and to compare it to the current beliefs expressed in well-known empirical models.

For example, Miami model (Lieth, 1975; Esser, 1984) relates NPP either to mean annual temperature (T_a) or to annual precipitation (P_a):

$$\text{NPP}(T_a, P_a) = \min\{\text{NPP}_T(T_a), \text{NPP}_h(P_a)\}; \quad (1)$$

$$\text{NPP}_T(T_a) = \frac{1350}{1 + e^{1.315 - 0.119T_a}};$$

$$\text{NPP}_h(P_a) = 1350(1 - e^{-0.000664P_a})$$

where T_a and P_a are mean annual temperature and annual amount of precipitation, respectively.

Plotting NPP against T_a we can see (Fig. 1) that data fits well $\text{NPP}_T(T_a)$ in case of biomes of sufficient water supply. Moreover, in line with the underlying logic of Miami model. NPP of the biomes of limited water supply is lower than predicted by $\text{NPP}_T(T_a)$. Plotting NPP against P_a we can see (Fig. 2) that the curve of humidity dependence $\text{NPP}_h(P_a)$ runs across the typical NPP values of these biomes, whereas NPP of biomes of limited heat supply is lower than predicted by $\text{NPP}_h(P_a)$.

The reference data set also fits the Montreal model (Box, 1988). This model supposes that NPP is controlled by actual evapotranspiration (AET):

$$\text{NPP}(\text{AET}) = 1350(1 - e^{-0.0009695(\text{AET} - 20)}). \quad (2)$$

The form of AET dependence suggested by formula Eq. (2) can be seen from the data (Fig. 3), although the biome of evergreen needle-leaf forests (36) drops out from the projected pattern.

Thus, plotting NPP against the climate characteristics we can see that the reference data set is consistent with the current beliefs about relation-

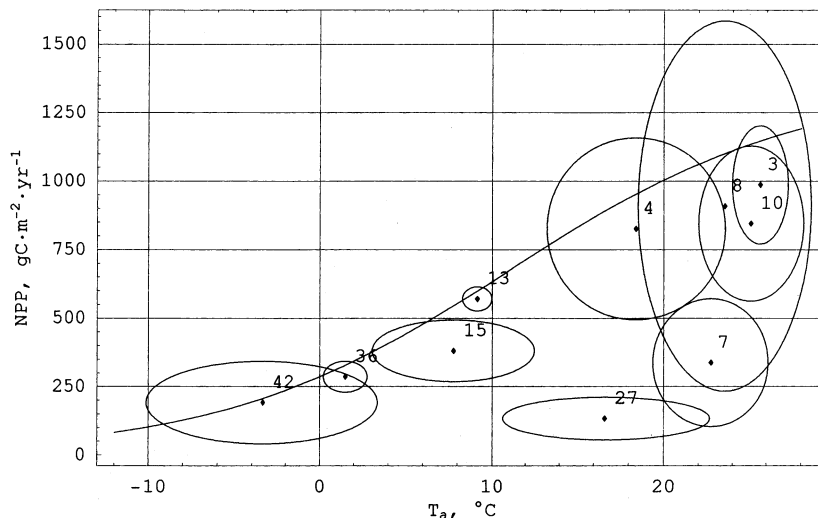


Fig. 1. NPP data plotted against mean annual temperature (T_a). Line represents the temperature dependence of NPP as suggested by the Miami model, formula Eq. (1). Diamonds mark the mean values of NPP and T_a of the following biomes: semi-desert scrubs (27), shrublands (7), subhumid woodlands (10), grasslands (15), tundra (42), needle-leaf forests (36), summer-green broad-leaved forests (13), evergreen broad-leaved forests (4), tropical rainforests (8), raingreen forests (3). Ellipses delineate confidence intervals of the mean values.

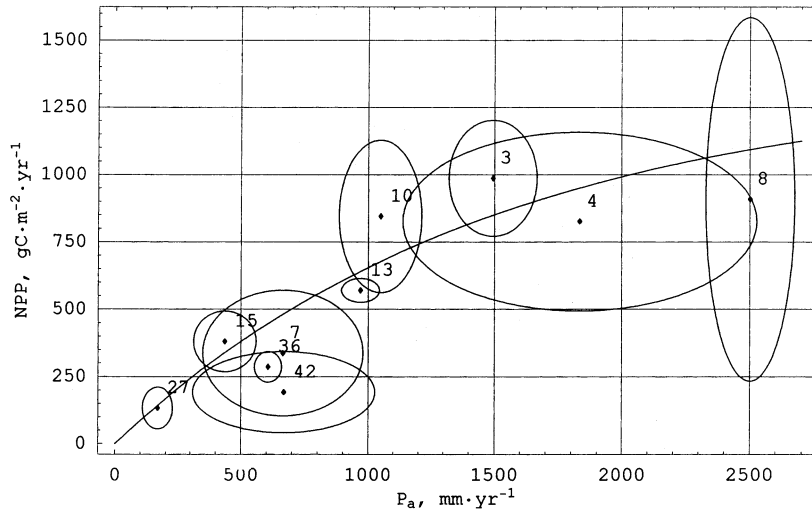


Fig. 2. NPP data plotted against annual precipitation (P_a). Line represents humidity curve of the Miami model. Diamonds mark the mean values of NPP and P_a of the following biomes: semi-desert scrubs (27), shrublands (7), subhumid woodlands (10), grasslands (15), tundra (42), needle-leaf forests (36), summer-green broad-leaved forests (13), evergreen broad-leaved forests (4), tropical rainforests (8), raingreen forests (3). Ellipses delineate confidence intervals of the mean values.

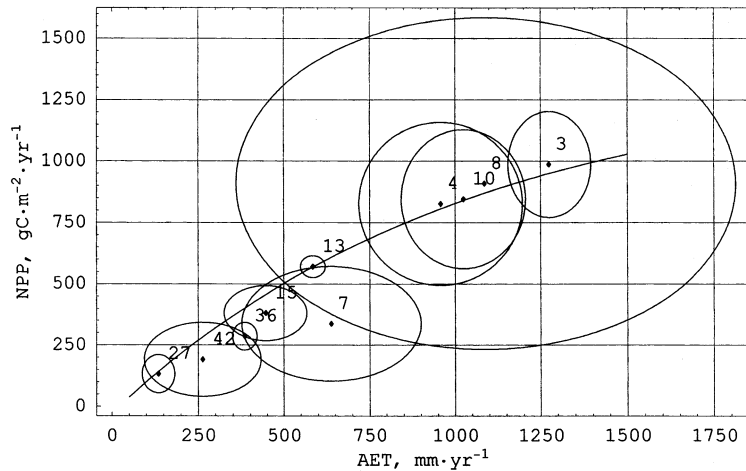


Fig. 3. NPP data plotted against actual evapotranspiration (AET). Line represents the Montreal model. Diamonds mark the mean values of NPP and P_a of the following biomes: semi-desert scrubs (27), shrublands (7), subhumid woodlands (10), grasslands (15), tundra (42), needle-leaf forests (36), summer-green broad-leaved forests (13), evergreen broad-leaved forests (4), tropical rainforests (8), raingreen forests (3). Ellipses delineate confidence intervals of the mean values.

ship between NPP and climate expressed in well-known empirical models.

2.1.2. Data interpretation

The empirical models considered above explain well inter-biome gradations of NPP by climate.

However, this is not to say that they explain within-biome gradations. By grouping data on the biome basis we take into account the fact that different species may show different NPP in the same climate. Then, we assume that species composition of a biome correlates in its turn to the

climate and formulate NPP distribution over the land as a function of the climate variables. In other words, similar to the gradient analysis of vegetation (Whittaker, 1975), we assume that vegetation is gradually changing along the gradient of a climate factor.

The idea that climate and species factors of NPP can be lumped together is not a new one. This assumption apparently underlies most of the empirical NPP models developed in 1970–80s. Thus, Chikugo model (Uchijima and Seino, 1985) explains the inter-biome gradations of NPP by Radiative Dryness Index and Net Radiation, that were proposed by Grigoriev and Budyko (Budyko, 1986; pp. 91–95) for predicting geographic locations of vegetation types. The input variables of Miami model, mean annual temperature and precipitation, were proposed by Volobuev (1964) for predicting geographic locations of major soil types.

Nevertheless, it is worth to mention that the use of biome-averaged values for model calibration restricts the area of model application to so-called *potential natural vegetation* (PNV), which is defined (Box, 1995) as the stable, terminal vegetation type which could occur on the area if all human disturbances were removed.

2.2. Restoring model parameters from the NPP data

Most of measurements are indirect. Temperatures are measured by changes in the height of a mercury column, weights are measured by changes in the length of a spring, and therefore it would be natural to consider NPP data as indirect measurements of model parameters.

A process-based model relates NPP to some driving factors at given values of parameters. Numerical inversion of the model relates parameters to the measured values of NPP and its driving factors. Let us consider, for example, a model that relates NPP directly to the amount of solar radiation (APAR) absorbed by canopy: $NPP(APAR) = \varepsilon APAR$, where ε is some kind of light-use efficiency (e.g. Ruimy et al., 1999). Inversion gives $\varepsilon(NPP, APAR) = NPP/APAR$ that allows us to restore ε from measured values of NPP and APAR and to consider the direct measurements of NPP and

APAR as indirect measurements of the light-use efficiency.

If the model would have more than one parameter, the inversion would not give us the values of each parameter, but rather the value of their linear combination. Because of this the number of model parameters must be reduced to a single lumped parameter (SLP).

The large number of parameters is an intrinsic feature of process-based models. This makes them applicable beyond the domain of available observations, where empirical models cannot be applied without serious doubts. Therefore, reduction in the number of model parameters should, nevertheless, conserve conceptual realism.

This goal can be achieved by taking into account self-organization of ecosystems and by use of ‘a unified theory of organization’ (Patten, 1995) emerging out of the various ecosystem studies. If we take into account self-organization we may assume that model parameters cannot vary independently, and that there are certain relationships between them. Assuming that some goal function, e.g. exergy (Jørgensen, 1992), is maximized (minimized) in course of ecosystem development, we may quantify, in some instances, the relationships between model parameters.

2.3. From point data to global field

If we look at the canopy functioning from the viewpoint of system theory, then climatic factors (including solar irradiance) are input variables, NPP is an output variable, and vegetation type is a state variable. The same input may result in different output depending on the system state. Describing behavior of this system by a model with a SLP, we thereby assume that SLP may serve as a quantitative characteristic of vegetation.

We may assume further that SLP, like vegetation, may be gradually changing along the gradient of a climate factor. For seeing the shape of the curve describing its climate dependence we may plot SLP against climate characteristics. Then proceeding from their biome-averaged values (i.e. averaged over the data points belonging to the same biome) we may formulate an empirical model linking SLP and some climatic variables.

Since geographical distribution of climatic variables is known, this empirical model enables us to assign SLP to each node of the geographical grid. We need just apply this model over the available climate fields (Leemans and Cramer, 1991) for this purpose.

All the known maps of terrestrial NPP were derived from irregularly scattered data. The scheme described above (Fig. 8) is a further method for interpolating in situ NPP measurements over geographical grid. It is close to the method of ‘geographic modelling’ (Box and Meentemeyer, 1991). As in case of ‘geographical modelling’ we use an empirical model linking the variable under interest with some climatic variables of known geographical distribution. However, methods differ in the choice of the function describing the relationship between NPP and climate. In case of ‘geographic modelling’ it is chosen arbitrary. We suggest that it should be derived from a process-based model. Obviously, process-based models are not all suitable for this purpose, but some of them are. An example of such a model we consider below.

3. Application of the scheme: TsuBiMo and its performance

The starting point of any process-based model of NPP is the solar energy input. The amount of solar radiation absorbed during the growing season determines the maximum yield of photosynthesis (Y_v) that could be obtained over the growing season:

$$Y_v = c_1 c_2 G D k \beta I_0, \quad (3)$$

where G is length of growing season, I_0 is light intensity at noon, D is average day length during growing season, β is light-use efficiency (pre-set at 2.0 mg CO₂/dm²/h per klux), k is the constant for conversion assimilated CO₂ to synthesized dry matter ($k = 0.6$ (mg dw)/(mg CO₂)), c_1 is the coefficient for expressing the stock of dry matter in carbon units ($c_1 = 0.45$ (g C)/(g dw)), and c_2 is the coefficient for coming from (mg dw)/dm² to (g dw)/m² ($c_2 = 0.1$ [(g dw)/m²]/[(mg dw)/dm²]).

Gross Primary Production (GPP) forms only a small fraction of Y_v :

$$GPP = Y_v S_f, \quad (4)$$

where S_f is the ratio of the realized GPP to the potential yield of photosynthesis.

Assuming (Box, 1988) that

$$NPP = 1350(1 - \exp(-GPP/1863)), \quad (5)$$

where NPP and GPP are given in g C/m² per year, we calculated S_f for each data point and found that it increases with T_v (Fig. 4) for the biomes with sufficient water supply.

The pattern of temperature dependence fits well the bell-shaped curve:

$$S_{f,T}(T_v) = 0.1532e^{-((T_v - 25)/12.8538)^2}. \quad (6)$$

Characterizing the effect of water limitation via rain factor ($RFL_v = P_v/T_v$), we plotted $S_f/S_{f,T}(T_v)$ against RFL_v and found that pattern of the humidity factor ($H_s = S_f/S_{f,T}(T_v)$) fits well the sigma-shaped curve:

$$H_s(RFL_v) = \begin{cases} e^{((RFL_v - 5.2)/2.7)^2}, & 0 < RFL_v \leq 5.2 \\ 1, & RFL_v > 5.2 \end{cases} \quad (7)$$

hence, we may express S_f as a function of T_v and RFL_v .

$$S_f(T_v, RFL_v) = S_{f,T}(T_v) H_s(RFL_v). \quad (8)$$

Proceeding to the more detailed description of the process (Oikawa, 1986), we can see that S_f of a biome depends on the photosynthetic capacity of the predominant vegetation (p_K):

$$S_f = 2 \ln \left\{ \frac{1 + \sqrt{1 + S_I}}{1 + \sqrt{1 + \phi S_I}} \right\} S_I^{-1}; \quad S_I = \frac{\beta I_0}{p_K}, \quad (9)$$

where $\phi = \exp(-K LAI)$, $p_K = p_{\max}/K$, $p_{\max}$ is light-saturated rate of photosynthesis and K is light extinction coefficient, LAI is leaf area index.

In order to reduce the number of variables, we assumed that the mode of foliage distribution is changing in such a manner that K and LAI are in a specific correspondence $K_{\text{opt}}(LAI)$ which gives a maximum of GPP for a given LAI (Alexandrov and Oikawa, 1997). This relationship implies minor variations in FPAR (fraction of absorbed

photosynthetically active radiation) of continuous vegetation cover, and so we set φ at 0.1. This allowed us to solve Eq. (10) with respect to p_K for

each data point and found that p_K , in its turn, depends on the same climate factors as S_f and in roughly similar manner (Figs. 5 and 6):

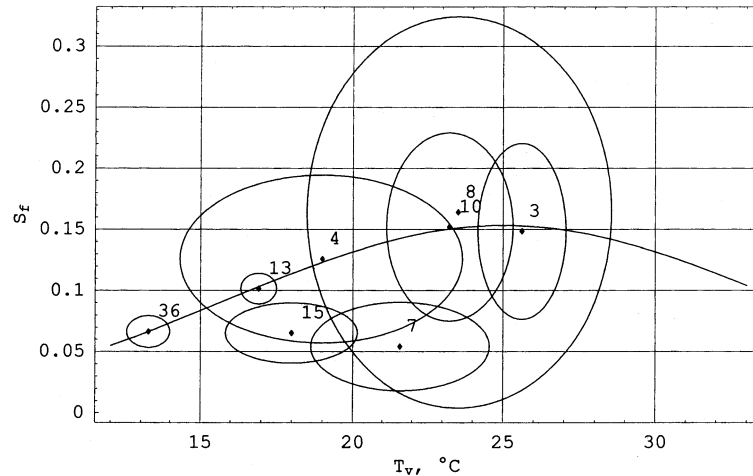


Fig. 4. The fraction of the potential yield of photosynthesis realized as GPP (S_f) as derived from the data by use of formulas Eqs. (3)–(5) plotted against the temperature of growing season (T_v). Line represents formula Eq. (6). Diamonds mark the mean values of S_f and T_v of the following biomes: semi-desert scrubs (27), shrublands (7), subhumid woodlands (10), grasslands (15), tundra (42), needle-leaf forests (36), summer-green broad-leaved forests (13), evergreen broad-leaved forests (4), tropical rainforests (8), raingreen forests (3). Ellipses delineate confidence intervals of the mean values.

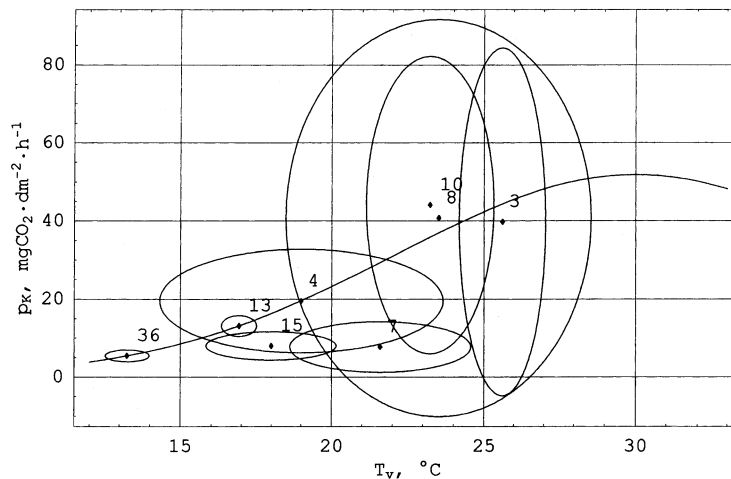


Fig. 5. Light saturated rate of photosynthesis divided by light attenuation coefficient (p_K) plotted against the temperature of growing season (T_v). Line represents formula Eq. (10). Diamonds mark the mean values of p_K derived from the data and T_v of the following biomes: semi-desert scrubs (27), shrublands (7), subhumid woodlands (10), grasslands (15), tundra (42), needle-leaf forests (36), summer-green broad-leaved forests (13), evergreen broad-leaved forests (4), tropical rainforests (8), raingreen forests (3). Ellipses delineate confidence intervals of the mean values.

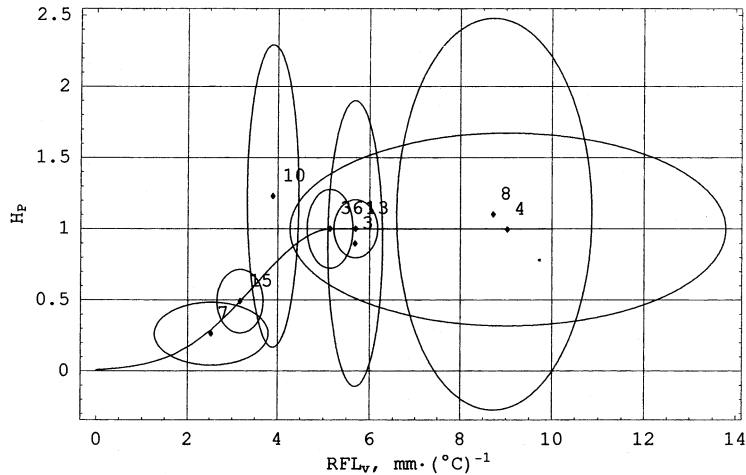


Fig. 6. Humidity factor of p_K plotted against rain factor (RFL_v). Line represents formula (11). Diamonds mark the mean values of $p_K p_{K,T}(T_v)^{-1}$ and RFL_v of the following biomes: semi-desert scrubs (27), shrublands (7), subhumid woodlands (10), grasslands (15), tundra (42), needle-leaf forests (36), summer-green broad-leaved forests (13), evergreen broad-leaved forests (4), tropical rainforests (8), raingreen forests (3). Ellipses delineate confidence intervals of the mean values.

$$p_{K,T}(T_v) = 51.8e^{-((T_v - 30)/11.1651)^2}, \quad (10)$$

$$H_p(RFL_v) = \begin{cases} e^{((RFL_v - 5.2)/2.4)^2}, & 0 < RFL_v \leq 5.2 \\ 1, & RFL_v > 5.2, \end{cases} \quad (11)$$

$$p_K(T_v, RFL_v) = p_{K,T}(T_v)H_p(RFL_v). \quad (12)$$

The model described by Eqs. (3)–(5) and Eqs. (9)–(12) will be referred to as *TsuBiMo*.

3.1. Predictive ability of the *TsuBiMo*

The ability of a process-based model to predict the NPP outside the domain of observations is normally assumed by its conceptual validity. However, a part of the *TsuBiMo* model, the climate dependence of p_K , is of empirical nature, and so the predictive ability of this sub-model can be evaluated only by comparing predictions to data. For this purpose one may apply the procedure which is referred to as data splitting.

The bell-shaped curve that we used to describe temperature dependence of p_K :

$$p_{K,T}(T_v) = p_{K,max} \cdot e^{-\frac{(T_v - T_{v,opt})^2}{\sigma^2}} \quad (13)$$

is completely determined by the two points $\{T_{v,13}, p_{K,13}\}$ and $\{T_{v,36}, p_{K,36}\}$ for the given $T_{v,opt}$:

$$\sigma^2 = \frac{(T_{v,36} - T_{v,opt})^2 - (T_{v,13} - T_{v,opt})^2}{\ln(p_{K,13}/p_{K,36})}, \quad (14)$$

$$p_{K,max} = p_{K,13}e^{((T_{v,13} - T_{v,opt})/\sigma)^2}$$

where $T_{v,13}$, $T_{v,36}$, $p_{K,13}$, and $p_{K,36}$ are the mean T_v and mean p_K of the deciduous broad-leaf forest (13) and evergreen needle-leaf forest (36), respectively.

This curve calibrated by the data from these two biomes predicts the p_K of the other biomes.

The branch of the bell-shaped curve used to describe the pattern humidity factor is determined by the point $\{RFL_{v,15}, H_{p,15}\}$, as the height of the bell is pre-set at 1:

$$\sigma^2 = \frac{(RFL_{v,15} - RFL_{v,opt})^2}{\ln(1/H_{p,15})}, \quad (15)$$

where $RFL_{v,15}$ and $H_{p,15}$ are RFL_v and H_p of the grasslands (15), respectively.

Hence, $H_p(RFL_v)$ calibrated for the grasslands predicts well H_p of the shrublands (7)

3.2. Evaluating *TsuBiMo* behavior at global scale

Applying *TsuBiMo* to the global fields of G , D , I_0 , T_v , and P_v , we face the problem posed by Cramer and Field (1999): there is no way in which the model behavior at global scale can be evalu-

ated except by comparison to other models of this sort. Therefore, we compare average NPP of the major biomes estimated by means of TsuBiMo with those obtained by means of the Miami and Montreal models (Table 1).

This comparison shows us some relations between model behavior at global scale and its calibration. Being applied to the region of evergreen needle-leaf forests, Montreal model gives significantly higher estimate of NPP_{36} than Miami model and TsuBiMo (Table 1), and Miami model gives here slightly higher estimate than TsuBiMo. Plotting model projections against the data, we notice the same order in the distance between model estimates of NPP_{36} and its value suggested by data: TsuBiMo fits the data closely (Fig. 5), Miami slightly overestimates NPP_{36} (Fig. 1), and Montreal model significantly overestimates it (Fig. 3). In other words, some discrepancies between the model projections can be traced from the calibration plots to global maps and attributed to calibration biases.

Not all discrepancies, however, can be easily attributed to calibration biases. In case of deciduous broad-leaved forests (13), all three models fit the data with a good precision, but differ in the regional value. Discrepancies of this sort may arise from the lack of correspondence between annual characteristics of climate and those of growing season. The best example is the biome of larch forests (14). Proceeding from the fact that T_a of the larch forests region is lower than that of tundra, Miami model suggests that NPP here is also lower than that of tundra (42). TsuBiMo suggests the opposite, for T_v of larch forests is higher than that of tundra. Relating productivity of larch forests to mean annual temperature is apparently a conceptual bias: mean annual temperature is not an appropriate characteristic of growth conditions in this case.

3.3. Robustness of inferences

Some discrepancies between models may stem from the differences in the process description. For example, the short version of TsuBiMo given by formulas Eqs. (3)–(8) fits the data on needle-leaf forests (36) and summer-green forests (13)

with the same precision as the full version, but gives slightly different regional estimates (Table 1). Which of these estimates is more credible? One may assume that the more sophisticated model provides the more credible result, but how to support this assumption by data.

If the discrepancies between compared models cannot be attributed either to calibration or conceptual biases, then none of them can be considered as the best. In that case we should characterize the uncertainty associated with the model choice. Showing the estimates derived from the same data by use of different conceptual scheme suits well for this purpose.

In order to characterize the uncertainty associated with the model choice let us consider the conceptual scheme proposed by Monteith (1972) and used in a number of NPP models (Maisongrande et al., 1995; Knorr and Heimann, 1995; Potter et al., 1993). This scheme relates NPP directly to the amount of solar radiation absorbed by plant canopy and leads us to the following model (see Ruimy et al., 1999 for the review of the relationships between APAR and NPP):

$$NPP_R(APAR) = 0.3675(APAR - 231), \quad (16)$$

$$H_{NPP}(RFL_v) = \begin{cases} e^{-((RFL_v - 5.2)/3.7)^2}, & 0 < RFL_v \leq 5.2 \\ 1, & RFL_v > 5.2, \end{cases} \quad (17)$$

$$NPP(APAR, RFL_v) = NPP_R(APAR)H_{NPP}(RFL_v), \quad (18)$$

where APAR is characterized by the amount of photosynthetically active radiation incoming during growing season, in MJ/m² per year.

Since the data fit this model surprisingly well (Fig. 7), we included in Table 1 the estimates derived by use of this scheme to show the robustness of the inferences derived from the reference data set.

4. Discussion

One should not expect too much of a global scale model. Actual NPP at a given site depends on the factors of local importance that is hardly

Table 1
NPP of major biomes (in g C/m² per year) estimated by global scale models based on the Osnabrück collection of NPP data

<i>Biome</i>	Miami model, Eq. (1)	Montreal model, Eq. (2)	TsuBiMo full, Eqs. (3)–(5), (9)–(12)	TsuBiMo short, Eqs. (3)–(8)	TsuBiMo direct, Eqs. (16)–(18)	Mean
Code	Name					
3	Raingreen forests	827	862	1084	1011	949
4	Evergreen broad-leaved forests	811	831	972	913	875
6	Deserts	66	35	48	54	115
7	Shrublands	368	460	416	382	381
8	Tropical rainforests	1040	973	1128	1027	948
10	Subhumid woodlands	635	747	909	834	791
13	Summer-green broad-leaved forests	522	544	570	538	505
14	Larch forests	96	219	182	200	234
15	Grasslands	404	498	504	469	428
27	Semi-desert scrubs	232	264	116	104	107
36	Needle-leaf forests	251	347	240	248	289
42	Tundra	106	191	68	90	147
						947 ± 92
						880 ± 57
						64 ± 27
						401 ± 33
						1023 ± 61
						783 ± 90
						536 ± 22
						186 ± 48
						460 ± 38
						165 ± 67
						275 ± 39
						120 ± 43

visible at the global scale, and thus no global scale model can predict it with confidence. Therefore, the target of a global scale model is regional NPP, that is, NPP of vegetation zones, biomes, and other regions of more or less uniform vegetation, soil and climate.

The bulk of NPP measurements makes it possible to validate a global scale model. Plotting biome-averaged NPP against its driving factor allows us to see the shape of the empirical relationship and the goodness-of-fit provided by the model. The only problem is that the available data do not constrain models to fit a single curve. Confidence intervals for the estimates of biome-averaged NPP are wide (Figs. 1–3), and hence most of models would be equally well validated by these data. In other words the data allow us to validate a model, but they hardly allow us to discriminate the best model from the worse one.

A complex process-based model has certain advantage over an empirical one: it is more flexible and thus more suitable for predicting NPP in changing environment. However, in the lack of complex data, the model complexity becomes a drawback. Due to limited understanding of many ecophysiological processes there is no way for

justifying any specific setting of some model parameters other than by fitting model output to the observed NPP values; and in its turn employing goodness-of-fit as a criterion for model validity penalizes complicated model structure, for multiple parameter calibration is unlikely to be free from personal prejudices. Reduction in the number of model parameters generally leads to a less complicated model with more objective parameter values, and reduction to a single parameter make its value to be completely determined by data.

The trade-off between complexity and objectivity becomes the issue when a process-based model is becoming recognized as decision-aid tool (Larocque, 1999). Many of NPP models which are used with this purpose in mind (Valentine, 1999; Peng and Apps, 1999; Goetz et al., 1999) might be converted to a single-parameter form and validated for the use at the global scale in the same manner (Fig. 8) as we validated TsuBiMo. The question is how to compensate the loss in complexity that would be incurred by such a conversion. Reichert and Omlin (1997) argue that prejudice against complex but poorly validated models can lead to an underestimation of the uncertainty in model predictions associated with the model structure. We suggest that one can

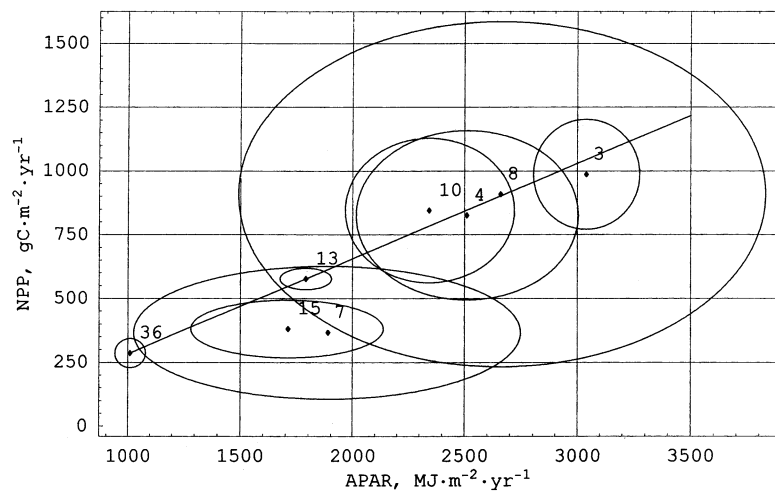


Fig. 7. NPP data and plotted against APAR. Line represents formula (16). Diamonds mark the mean values of NPP and AET of the following biomes: semi-desert scrubs (27), shrublands (7), subhumid woodlands (10), grasslands (15), tundra (42), needle-leaf forests (36), summer-green broad-leaved forests (13), evergreen broad-leaved forests (4), tropical rainforests (8), raingreen forests (3). Ellipses delineate confidence intervals of the mean values.

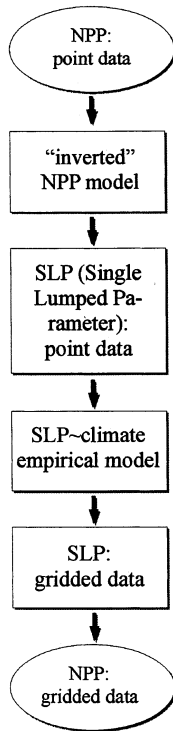


Fig. 8. The frame diagram of the proposed scheme for globalization of a process-based NPP model.

avoid such pitfalls by comparing predictions inferred by different models from the same data. In other words we assume that in the search of optimal trade-off between complexity and objectivity, we may supersede a single model by an ensemble of different models.

From the practical point of view, formation of the ensemble of accurately validated NPP models is more urgent task than further improvement of any of them. Inclusion of terrestrial carbon sinks into the scope of international agreements on CO₂ emission reduction calls for consensual knowledge of carbon sinks functioning. However, consensual knowledge need not be expressed in the form of a single ‘true’ model. There are a variety of models that fit equally well to available data but differ in predictions over the global scale. Therefore, it seems to be more realistic to express consensual knowledge as a statistical description of a large collection of identically calibrated models.

5. Conclusion

It is more or less obvious that a process-based NPP model tested for a stand can be applied over the global grid, but doing so, we face the problem—how to calibrate it at this scale. Our approach to this problem is as follows. In the first place, we considered the data on NPP as indirect measurement of model parameters. Secondly, we reduced the number of undefined parameters to SLP. In the third place, we ‘inverted’ the model and derived the value of the SLP for each data point. In the fourth place, we considered the SLP as some characteristic of vegetation and formulated empirical model linking SLP and climate averaged over the data points belonging to the same biome. And finally, we globalized the process-based model by assigning SLP value to each node of the geographical grid proceeding from the climate characteristics of the node. Applying this scheme to the process-based model proposed by Oikawa (1986) we derived a global scale model of NPP (named as TsuBiMo) from the Osnabrück collection of NPP data (Esser et al., 1997) or, to be more precise, from certain subset (Alexandrov et al., 1999) of this collection.

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Appendix A. Nomenclature

AET	Actual evapotranspiration (mm per year)
APAR	Absorbed photosynthetically active radiation (MJ/m ² per year)
c_1	The coefficient for expressing the stock of dry matter in carbon units ((g C)/(g dw))

c_2	The coefficient for coming from (mg dw)/dm ² to (g dw)/m ² ((g dw/m ²)/(mg dw/dm ²))
D	Day length averaged over the growth period (hours)
G	The length of growth period (day)
GPP	Gross primary production (g C/m ² per year)
H_S, H_p	Humidity factor of the climate dependence of S_f and p_K , respectively
I_0	Light intensity at noon averaged over the growth period (klux)
k	Conversion factor for coming from mg CO ₂ to mg of dry matter ((mg dw)/(mg CO ₂))
K	Light extinction coefficient
LAI	Leaf area index (dm ² /dm ²)
NPP	Net primary production (g C/m ² per year)
P_a	Annual precipitation (mm)
P_v	Mean monthly precipitation of growth period (mm)
$p_{\max}$	Light-saturated rate of photosynthesis ((mg CO ₂)/dm ² per hour)
p_K	$p_{\max}$ divided by K ((mg CO ₂)/dm ² per hour)
RFL _v	Rain factor of growth period (P_v divided by T_v)
S_f	The fraction of the potential yield of photosynthesis realized as GPP
T_v	Mean monthly temperature of growth period (°C)
T_a	Mean annual temperature (°C)
Y_v	Potential yield of photosynthesis (g C /m ² per year)
β	Light use efficiency of photosynthesis ((mg CO ₂)/dm ² /h klux)
ϕ	Fraction of solar radiation which is not absorbed by canopy

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